

Database Administration

Course Title: Database Administration
Course No: BIT352
Nature of the Course: Theory + Lab
Semester: VI

Full Marks: 60+ 20+20
Pass Marks: 24+8+8
Credit Hrs: 3

Course Description:

This course introduces the database administration techniques in Oracle. Most of the DBA's tasks are covered. Topics covered include principles of DBA Roles, Oracle Database Architecture and storage management, Database backup, restoration and recovery, connectivity and user management for database security, Tuning of database and overall DB administration which could be useful for administrator in the future.

Course Objectives:

To understand the basic role, task and responsibilities of Database Administrator, understand the Oracle database architecture, be able to install and configure an Oracle Database, be able to create database objects like tables, views, indexes etc. and able to write PL/SQL Procedures, be able to administer the Oracle Database, create and manage storage structures, create and manage the users, be able to perform backup and recovery, tuning the oracle database for the better performance.

Course Contents:

Unit 1: Introduction Administration Basics (7Hrs.)

- 1.1. Introduction: Overview of the Oracle Database, Relational Database Concepts, Overview of Oracle Database Architecture, Overview of Oracle Multitenant Architecture, Overview of Oracle Data Guard, Oracle RAC, Oracle ASM, Oracle Cloud, Database Administration Tasks.
- 1.2. Oracle Architecture: Oracle Database Architecture, the Oracle Database Instance, Memory Structures, Process Architecture and Structures, Server and Client Processes, Database Storage Architecture, Connect to the Oracle Database Instance.
- 1.3. Administer Database Instance: Administrative Tools Available to a DBA, Use SQL*Plus in Oracle to Manage a Database Instance, use SQL Developer to Manage a Database Instance, Administer the Database using Enterprise Manager (EM), Overview of the Enterprise Manager Framework, Enterprise Manager Cloud Control, Initialization Parameter Files

Unit 2: Network Environment and Storage Structures (5 Hrs.)

- 2.1. Configuring Network Environment: Overview of Network Configuration, Oracle Net Listener Configuration and Management, Oracle Net Naming Methods, Tools for Configuring and Managing the Oracle Network, using the Net Configuration Assistant,

Configure Client Connections with Net Manager, View Listener Configuration, Start and Stop the Oracle Listener, use TNSPING to Test Oracle Net Connectivity, Connect to the Database, Configure Net Services with Enterprise Manager.

- 2.2. Storage Structures: Overview of Data Storage, the Database Block, Overview of Tablespaces and Datafiles, use Enterprise Manager to view the Storage Structure of the Database, Create and Alter Tablespace Commands, Temporary Tablespaces, Create and Manage Datafiles, use OMF, Drop Tablespaces and Datafiles

Unit 3: Users and Privileges (5 Hrs.)

- 3.1. Manage Users: Predefined Database Administration Accounts, User Accounts, Create a User Account, User Authentication, Change a User's Password, Manage a User Account, Drop a User Account, Monitor User Information, Terminate User Sessions, Create a Schema Only Account
- 3.2. Manage Privileges: Database Access, Oracle Supplied Roles, System and Object Level Privileges, the Grant and Revoke Commands, Create, Modify and Drop Roles, Use Predefined Roles

Unit 4: Profiles, Resources, and Auditing (4 Hrs.)

- 4.1. Profiles And Resources: Overview of User Profiles, Profile Resource Parameters, Create Profile Command, Manage Passwords with Profiles, Control Resource Usage with Profiles, Maintain Profiles
- 4.2. Database Auditing: Overview of Database Security, Overview of Database Auditing, Security Compliance, Standard Auditing, Unified Audit Trail, Separation of Audit Responsibilities with The AUDIT_ADMIN And AUDIT_VIEWER Roles, Configure the Audit Trail, Specify Audit Options.

Unit 5: Concurrency, Backup and Recovery (7 Hrs.)

- 5.1. Concurrency: Levels of Locking in Oracle, Methods Used to Acquire Locks, Data Concurrency, Possible Causes of Contention, DML Locks, Prevent Locking Problems, Detect Lock Contention, Resolve Conflicts
- 5.2. Undo Management: Undo Data Overview, Monitor and Administer Undo, Configure Undo Retention, Switch Undo Tablespaces, Specify the Retention Period, Guarantee Undo Retention, Retention Period for Flashback Queries, View Undo Space Information, use the Undo Advisor, Size the Undo Tablespace, Alter an Undo Tablespace to a Fixed Size
- 5.3. Backup And Recovery: Oracle Backup Solutions, Oracle Suggested Backup Strategy, Overview of Database Backup, Restore and Recover, Flashback Technology, Types of Failure, Instance Recovery, Tune Instance Recovery, The MTTR Advisor, Media Failure, Configure a Database for Recoverability

Unit 6: Database Maintenance and Performance Management (4 Hrs.)

- 6.1. Database Maintenance: Overview, View the Alert Log, the Automatic Workload Repository, Statistic Levels, the Automatic Database Diagnostic Monitoring, Monitor an Oracle Database, use the Advisors, Set Up Notification Rules
- 6.2. Performance Management: Tuning Information Sources, Performance Monitoring, Tuning Activities, Performance Planning, Instance Tuning, Performance Tuning Methodology, Performance Tuning Data, Monitoring Performance, Managing Memory, Manage Private Temporary Tables

Unit 7: Tuning and Moving (5 Hrs.)

- 7.1. Tuning: SQL Tuning, Cancel a SQL Statement in a Session, the Oracle Optimizer, SQL Plan Directives, Adaptive Execution Plans, SQL Advisors, Automatic SQL Tuning Results, Implement Automatic Tuning Recommendations, SQL Tuning Advisor
- 7.2. Moving Data: Create Directory Objects, Data Pump Architecture, Data Pump Data Dictionary Views, Data Pump Interactive Mode, Data Pump API, use Data Pump to Export Data, Use Data Pump to Import Data, Overview of SQL Loader, Command Line Parameters, Record Filtering, Control File Keywords, Data files, SQL Loader Data Paths, External Tables

Unit 8: Managing Resources and Scheduling Tasks (4 Hrs.)

- 8.1 Managing Resources: Overview of the Database Resource Manager, use the Resource Manager, Create Resource Plans, the Default Maintenance Resource Manager Plan, Create Resource Plan Directives, Allocate Resources for Resource Plans
- 8.2 Scheduling Tasks: Introduction to The Scheduler, Access Rights, Scheduler Components and Workflow, Create a Job, Job Classes, use Time Based, Event Based Schedules, Create an Event Based Schedule

Unit 9: Space Management and Oracle Support (4 Hrs.)

- 9.1. Managing Space: Overview of Space Management, Block Space Management within Segments, Segment Types, Allocate Extents, Allocate Space, Row Chaining and Migration, Create Tables without Segments
- 9.2. Oracle Support: The Enterprise Manager Support Workbench, Register for Security Updates, Work with Oracle Support, My Oracle Support Integration, Log Service Requests, Manage Patches, Apply a Patch

Laboratory Works:

Student should prepare lab sheet for most of the units in the syllabus. They should practice, design database and implement database administration activities that demonstrates different concepts discussed in the class.

References:

1. Oracle Database 19c DBA By Examples: Installation and Administration, Ravindra Gupta, 2021.
2. <https://docs.oracle.com/en/database/oracle/oracle-database/19/administration.html>
3. Pro Oracle Database 18c Administration: Manage and Safeguard Your Organization's Data, Michelle Malcher and Darl Kuhn, Third Edition.
4. Oracle Database 12c DBA Handbook, Manage a Scalable, Secure Oracle Enterprise Database Environment, Bob Bryla.
5. Oracle DBA Mentor: Succeeding as an Oracle Database Administrator, Brian Peasland.